Agriculture: Strawberry Pest Management Guidelines

Pesticide Application Checklist

When planning for possible pesticide applications in an IPM program, review and complete this checklist to consider practices that minimize environmental and efficacy problems.

Choose a pesticide from the UC IPM Pest Management Guidelines for the target pest, considering:

- Impact on natural enemies and pollinators(r734900311.html).
- Potential for water quality problems using the UC IPM WaterTox database(/TOX/simplewatertox.html).
- Impact on aquatic invertebrates. For more information, see *Pesticide Choice*(http://anrcatalog.ucanr.edu/pdf/8161.pdf), (PDF) UC ANR Publication 8161.
- Chemical mode of action, if pesticide resistance is an issue. For more information, see
  - *Herbicide Resistance: Definition and Management Strategies(http://anrcatalog.ucanr.edu/pdf/8012.pdf)* (PDF), UC ANR Publication 8012 .
  - *Fungicide Resistance in Crop Pathogens: How Can It Be Managed?(https://www.frac.info/media/jtafagre/monograph-1.pdf)* (PDF)
- Endangered species that may be near your site. Find out using the Department of Pesticide Regulation's PRESCRIBE(http://www.cdpr.ca.gov/docs/endspec/prescint.htm) program.

Before an application

- Ensure that spray equipment is properly calibrated(/training/incorporating-calibration.html) to deliver the desired pesticide amount for optimal coverage.
- Use appropriate spray nozzles and pressure to minimize off-site movement of pesticides.
- Avoid spraying during these conditions to avoid off-site movement of pesticides.
  - Wind speed under 3 mph and over 10 mph
  - Temperature inversions
  - Just prior to rain or irrigation (unless it is an appropriate amount, such as when incorporating a soil-applied pesticide)
  - At tractor speeds over 2 mph
- Identify and take special care to protect sensitive areas (for example, waterways or riparian areas) surrounding your application site.
- Review and follow labeling for pesticide handling, personal protection equipment (PPE) requirements, storage, and disposal guidelines.
- Check and follow restricted entry intervals (REI) and preharvest intervals (PHI).

After an application:

- Record application date, pesticide used, rate, and location of application.
- Follow up to confirm that treatment was effective.

Consider water management practices that reduce pesticide movement off-site.

- Consult relevant publications: *Protecting Surface Water from Sediment-Associated Pesticides in Furrow-Irrigated Crops*(http://anrcatalog.ucanr.edu/pdf/8403.pdf), UC ANR Publication 8403 (PDF).

- Consult the Department of Pesticide Regulation Groundwater Protection Program (GWPA)(http://www.cdpr.ca.gov/docs/emon/grndwtr) website for pesticide information and mitigation measures.
- Install an irrigation recirculation or storage and reuse(/mitigation/water_reuse.html) system. Redesign inlets into tailwater ditches to reduce erosion. For more information, see these publications:
  - *Reducing Runoff from Irrigated Lands: Tailwater Return Systems*(http://anrcatalog.ucanr.edu/pdf/8225.pdf), UC ANR Publication 8225 (PDF).
  - *Reducing Runoff from Irrigated Lands: Storing Runoff from Winter Rains*(http://anrcatalog.ucanr.edu/pdf/8211.pdf), UC ANR Publication 8211 (PDF).
- Use drip rather than sprinkler or flood irrigation.
- Limit irrigation to amount required using soil moisture monitoring and evapotranspiration(/WEATHER/#WEATHERDATA) (ET). (For more information, see *Reducing Runoff from Irrigated Lands: Understanding Your Orchard's Water Requirements*(http://anrcatalog.ucanr.edu/pdf/8212.pdf), UC ANR Publication 8212 (PDF).)
- Consider using cover crops.
- Consider vegetative filter strips(/mitigation/veg_filtering.html) or ditches. (For more information, see *Vegetative Filter Strips*(http://anrcatalog.ucanr.edu/pdf/8195.pdf), UC ANR Publication 8195 (PDF)
- Use polyacrylamide (PAM) tablets in furrow irrigation systems to prevent off-site movement of sediments.

Consider practices that reduce air quality problems.

- When possible, reduce volatile organic compound (VOC) emissions(/mitigation/reducing_voc.html) by decreasing the amount of pesticide applied, choosing low-emission management methods, and avoiding fumigants and emulsifiable concentrate (EC) formulations.

For more about mitigating the effects of pesticides, see the Mitigation page(/mitigation).

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